SQL – PGP DSBA

New Wheels Sales

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DIKSHA PANDEY

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Business Context:

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers. New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Problem Statement:

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

Objective:

As a data scientist, you see that there is an array of questions that are being asked at the leadership level that needs to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Data Description:

You have data on the vehicles you sell: what are the make, model, and year? What is the price point? Data on your customers, such as where they live and payment methods, data on orders and shipments, such as when the order was shipped and received, what the after-sales feedback was, and so on.

The data dictionary (new_wheels_sales.txt), which has detailed descriptions of each column, and the dump file (new_wheels_dumpfile.sql), can be found in the same **data** folder.
The new_wheels_proj.rar file contains the data folder and the diagram folder which contains the ER diagram.

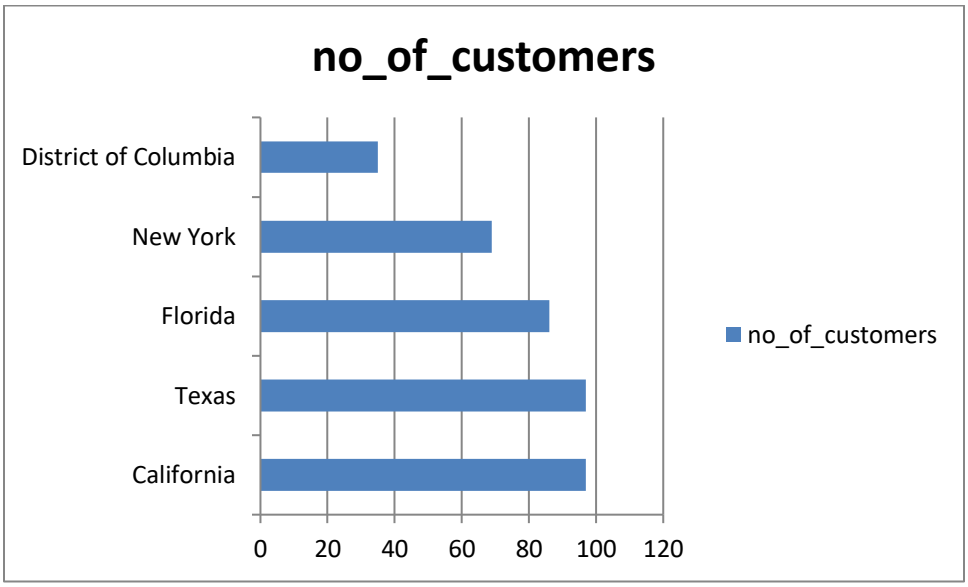
Business Overview

Title	Numbers
Total Revenue	**12.54cr**
Total Orders	**1000**
Total Customers	**994**
Avg. Rating	**3**
Last Qtr Revenue	**2.34cr**
Last Qtr Orders	**199**
Avg Days to Ship	**105**
% Good Feedback	**42%**

CUSTOMER METRICS

1. What is the distribution of customers across states?

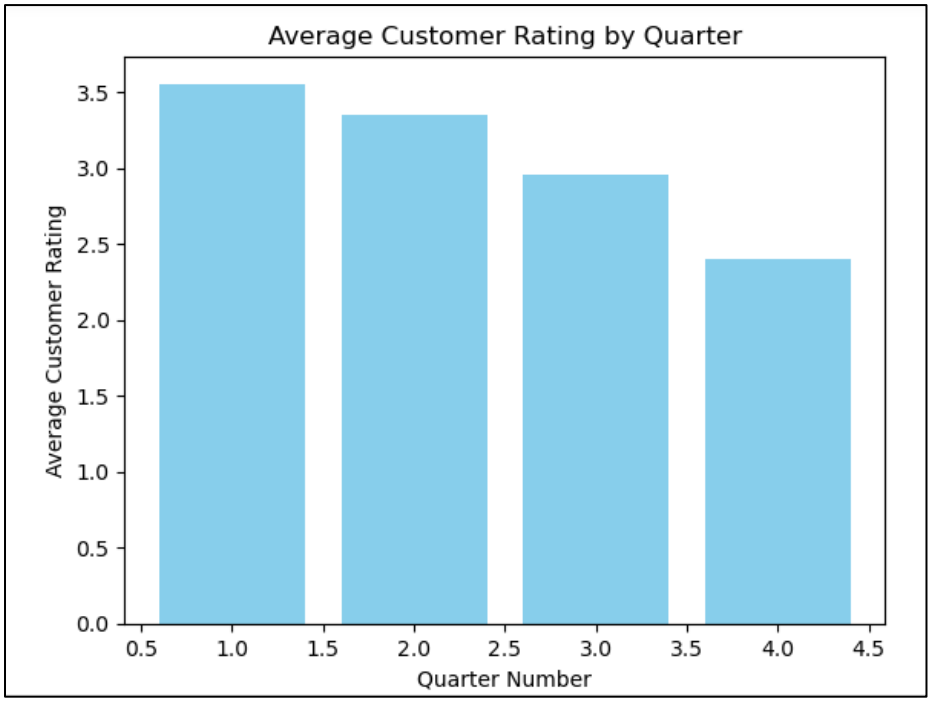
   Observation: The top 3 states observed are California, Texas, and Florida with highest number of customers.



state	no_of_customers
California	97
Texas	97
Florida	86
New York	69
District of Columbia	35

2. What is the average rating in each quarter?
   Very Bad is 1, Bad is 2, Okay is 3, Good is 4, Very Good is 5.

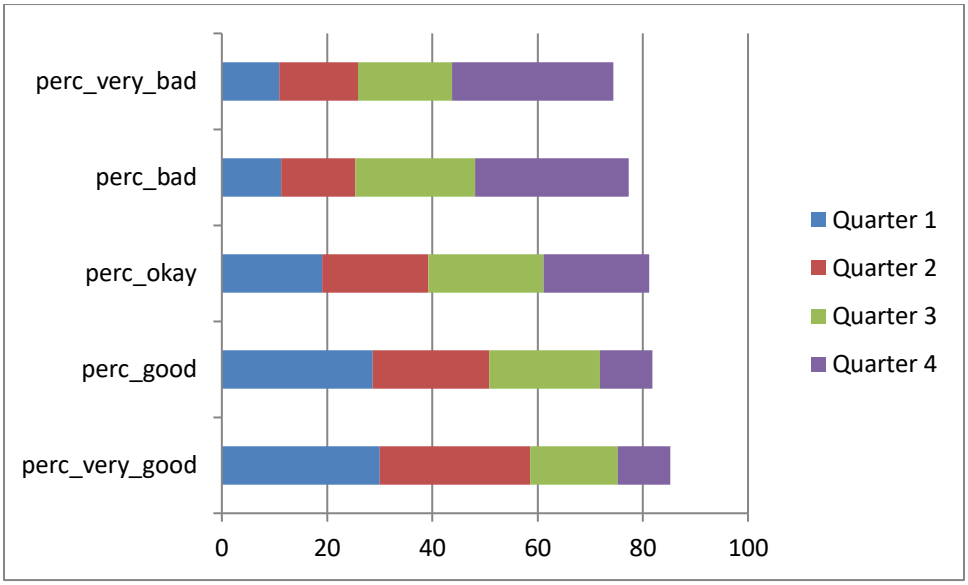
   <u>Observation:</u> We observe rating above 3 in the first two quarters, and below 3 in the 3rd and 4th quarter.



quarter_number	avg_feedback
1	3.5548
2	3.3550
3	2.9563
4	2.3970

3. Are customers getting more dissatisfied over time?

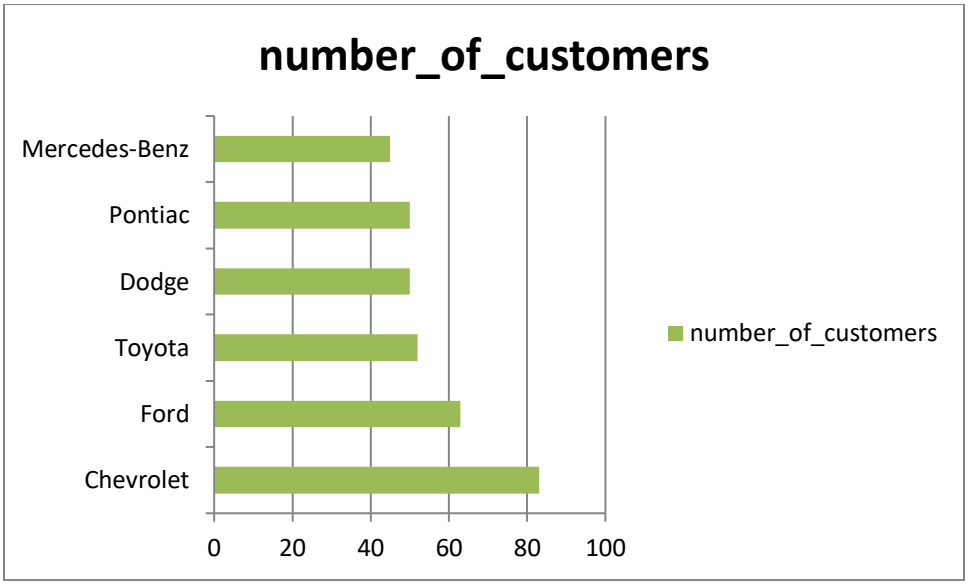
Observation: Yes, we see more dissatisfaction over a period of time, with each increasing quarters.



quarter_number	perc_very_good	perc_good	perc_okay	perc_bad	perc_very_bad
1	30.0000	28.7097	19.0323	11.2903	10.9677
2	28.6260	22.1374	20.2290	14.1221	14.8855
3	16.5939	20.9607	21.8341	22.7074	17.9039
4	10.0503	10.0503	20.1005	29.1457	30.6533

4. Which are the top 5 vehicle makers preferred by the customer?

Observation: The top three vehicle makers chosen by customers are Chevrolet, Ford and Toyota followed by Dodge and Pontiac.



vehicle_maker	number_of_customers
Chevrolet	83
Ford	63
Toyota	52
Dodge	50
Pontiac	50

5. What is the most preferred vehicle make in each state?

Observation: With the states given below and in the overall data we observe Chevrolet the most preferred, followed by Dodge, Pontiac, and Ford.

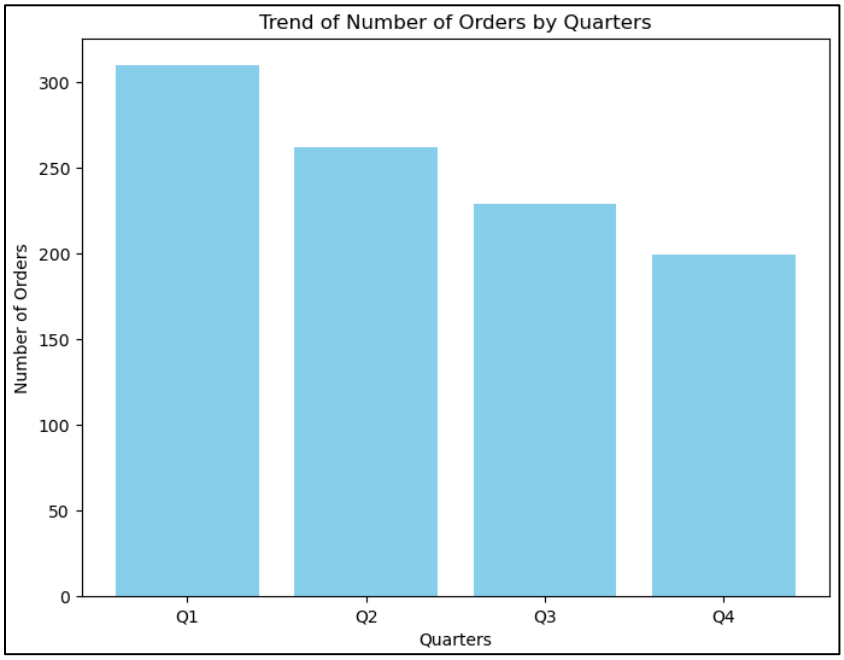
state	vehicle_maker	rnk
Alabama	Dodge	1
Alaska	Chevrolet	1
Arizona	Pontiac	1
Arizona	Cadillac	1
Arkansas	Suzuki	1
Arkansas	Chevrolet	1

Vehicle Maker	Count of state
Chevrolet	17
Dodge	12
Pontiac	11
Ford	10
Toyota	9
Mazda	8
Nissan	7

REVENUE METRICS

6. What is the trend of number of orders by quarters?

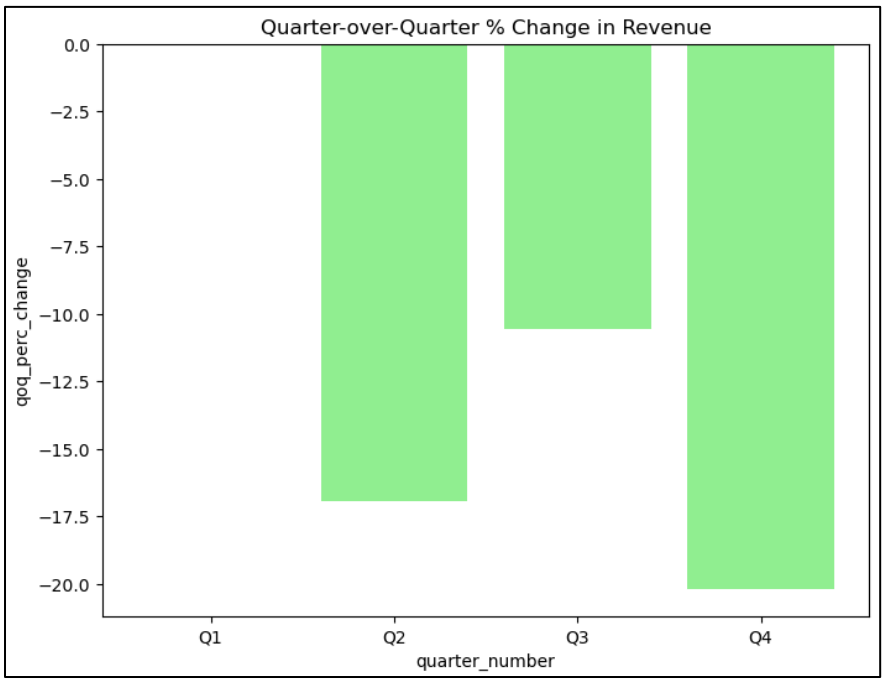
Observation: We notice a decrease in total orders towards quarter end.



quarter_number	total_orders
1	310
2	262
3	229
4	199

7. What is the quarter over quarter % change in revenue?

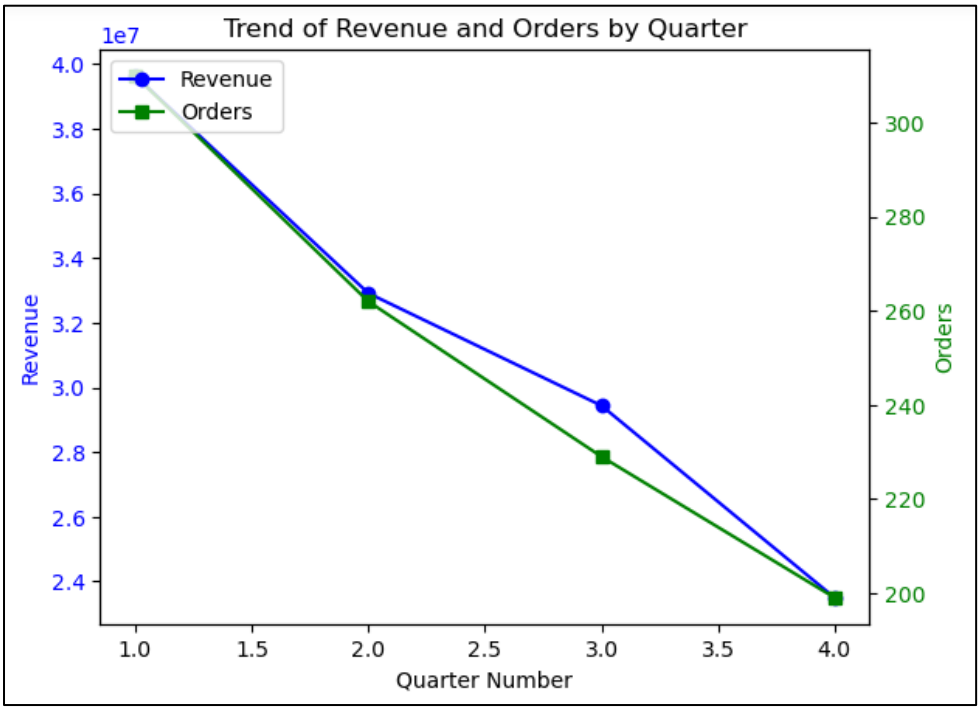
Observation: We observe a decrease in revenue quarter over quarter. The percentage revenue change is negative as seen below.



quarter_number	revenue	previous_revenue	qoq_perc_change
1	39637630.97	NULL	NULL
2	32913737.76	39637630.97	-16.963408
3	29435427.48	32913737.76	-10.567959
4	23496008.22	29435427.48	-20.177792

8. What is the trend of revenue and orders by quarters?

Observation: Order and Revenue found in a similar sloping down trend.

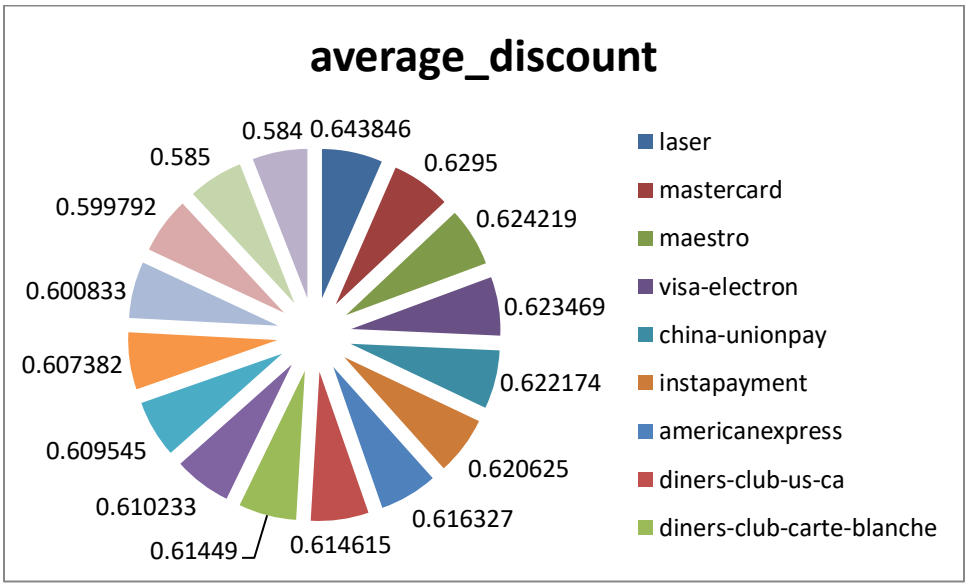


quarter_number	revenue	total_orders
1	39637630.97	310
2	32913737.76	262
3	29435427.48	229
4	23496008.22	199

SHIPPING METRICS

9. What is the average discount offered for different types of credit cards?

Observation: Average discount has been observed between 0.64 and 0.58 with highest for laser and master card.



credit_card_type	average_discount
laser	0.643846
mastercard	0.629500
maestro	0.624219
visa-electron	0.623469
china-unionpay	0.622174

10. What is the average time taken to ship the placed orders for each quarter?

Observation: Increase in average shipping time quarter by quarter.

quarter_number	average_shipping_time
1	57.1677
2	71.1107
3	117.7555
4	174.0955

Actionable Insights:

- **Customer Behavior Patterns:** Popular products brought are Chevrolet, Ford and Toyota. With an average order value of 83,115.
- **Sales Trends:** Downfall in sales trend observed. Affecting overall orders and revenue.
- **Operational Efficiency:** Increase in shipping time over each quarter has also affected overall decrease in orders hence affecting revenue and total sales.
- **Product Performance:** With an average discount of 0.61 provided.
- **Customer Satisfaction:** Customer feedbacks have deteriorated over each quarter. Hence affected the rating from very good to very bad.

Business Recommendation:

1. **Optimize Inventory Management:** Implement inventory management strategies to align stock levels with customer demand and sales trends. Utilize data-driven forecasting techniques to minimize stockouts, reduce excess inventory, and optimize warehouse space utilization.
2. **Enhance Customer Experience:** Invest in improving the overall customer experience by addressing common pain points identified in customer feedback and support interactions. Implement self-service options, streamline checkout processes, and provide personalized recommendations to enhance satisfaction and loyalty.
3. **Tailor Marketing Strategies:** Leverage customer segmentation and behavior analysis to tailor marketing campaigns and promotions to specific customer segments. Use targeted messaging, personalized offers, and omni channel marketing strategies to increase engagement and drive conversions.
4. **Optimize Pricing Strategies:** Analyze pricing data and competitive intelligence to optimize pricing strategies for maximum profitability and competitiveness. Consider dynamic pricing models, promotional pricing tactics, and value-based pricing strategies to maximize revenue and market share.
5. **Improve Operational Efficiency:** Identify inefficiencies in business processes and workflows and implement process improvements to enhance operational efficiency. Streamline order fulfillment, reduce lead times, and automate repetitive tasks to increase productivity and reduce costs.
6. **Expand Product Offerings:** Use insights from product performance analysis to identify opportunities for expanding product offerings or introducing new product lines. Conduct market research to identify emerging trends and customer preferences and develop products that meet evolving consumer needs.
7. **Strengthen Supply Chain Resilience:** Evaluate supply chain vulnerabilities and implement risk management strategies to mitigate potential disruptions. Diversify supplier partnerships, establish backup plans for critical components, and invest in supply chain visibility and transparency tools to enhance resilience.
8. **Invest in Technology:** Leverage technology solutions, such as data analytics platforms, customer relationship management (CRM) systems, and e-commerce platforms, to gain deeper insights,

improve decision-making, and drive innovation. Invest in digital transformation initiatives to stay competitive in the rapidly evolving business landscape.

9. **Focus on Sustainability:** Incorporate sustainability initiatives into business operations to meet growing consumer demand for environmentally friendly products and practices. Reduce carbon footprint, minimize waste, and adopt sustainable sourcing and manufacturing practices to attract eco-conscious customers and enhance brand reputation.
10. **Monitor Market Trends:** Stay informed about market trends, industry developments, and competitor activities to identify opportunities and threats. Monitor consumer behavior, economic indicators, and regulatory changes to adapt business strategies and maintain agility in response to evolving market conditions.